

Tegravance Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-274/B

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

No changes to the approved analytical procedures are proposed in this submission. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Cleaning verification swab results were below the health-based exposure limits derived for the product. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Control of Drug Product

The drug product specification includes tests for description, identification, assay, uniformity of dosage units, dissolution, related substances, water content, and microbial limits. Analytical procedures are compendial where available; non-compendial procedures were validated in accordance with ICH Q2(R1).

Cleaning verification swab results were below the health-based exposure limits derived for the product. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation.

3.2.P.4 Control of Excipients

For the commercial formulation F-274/B, Primojel (Sodium starch glycolate, disintegrant) is used.

The approach described herein is consistent with Ph. Eur. general monograph 2034, IPEC Excipient Qualification Guide and ICH Q8(R2).

Grade Selection Rationale

The selection of Primojel is justified by robust tablet hardness across the compression force design space observed during development studies.

Approved Drug Product Specification

The current approved commercial specification (SPEC-EX-8205 Rev 06) lists Sodium starch glycolate as Primojel.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

Batch Formula

The representative commercial batch formula is provided for a nominal batch size of 400,000 tablets. Quantities of all components are expressed per tablet and per batch. Purified water is used as granulation fluid and is removed during processing; it does not appear in the final composition.

Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Deviations observed during the campaign were investigated and closed with no impact to product quality. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

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performed per the site monitoring program, with alert and action limits established from historical data.

Process Validation Summary

Process performance qualification was executed on three consecutive commercial-scale batches. All predefined acceptance criteria for critical process parameters and critical quality attributes were met, supporting a conclusion of a validated state of control.

Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Cleaning verification swab results were below the health-based exposure limits derived for the product. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Facilities and Equipment

Manufacturing is performed at the approved commercial site using dedicated product-contact equipment trains. Equipment qualification and cleaning validation are maintained under the site validation master plan. No changes to the approved facility registration are proposed.

Cleaning verification swab results were below the health-based exposure limits derived for the product. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Deviations observed during the campaign were investigated and closed with no impact to product quality. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The control strategy links critical quality attributes to critical process parameters identified during development.

Stability Summary

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Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Deviations observed during the campaign were investigated and closed with no impact to product quality. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control.